

ABEKA SCHOLAR ACADEMY

Subj Grade 12 Pre Calculus / Calculus Curriculum

Official Curriculum & Study Material

Chapter 1

Welcome to our amazing journey into calculus / calculus! We have so many fun things to discover today. Let's get started! Did you know? Education is about learning skills and knowledge. It also means helping people to learn how to do things and support them to think about what they learn. It is also important for educators to teach ways to find and use information. Education needs research to find out how to make it better. Through education, the knowledge of society, country, and of the world is passed on from generation to generation. This may include education in morality, for example learning how to act as loyal, honest and effective citizen.

Here's a really cool secret about this topic: Education may help and guide individuals from one class to another. Educated people and groups can do things like help less-educated people and encourage them to get educated. There are different ways to categorize education: for example, by age or subject. One way is to divide it into formal education, non-formal education, and informal education. Formal education is usually in school, where a person may learn basic, academic, or trade skills. Small children often attend a nursery or kindergarten but often formal education begins in elementary school and continues with secondary school. Post-secondary education, or higher education, is usually at a college or university which may grant an academic degree.

Close your eyes and picture this... Or, students may go to a City college where they learn practical skills. This way learners can become qualified to be plumbers, electricians, builders and similar occupations. These course have arrangements for students to get practical experience. Apprenticeship was the older way to do this, Non-formal education includes adult basic education, adult literacy education or school equivalency preparation. In non-formal education, someone who is not in school can learn to read, write, and other basic life or job skills. Home education, individualized instruction, such as programmed learning, distance learning and computer-assisted instruction are other possibilities. Let me tell you a fascinating story. Informal education is less organized. It may be a parent (like a dad or mom) teaching a child how to make a meal or ride a bicycle. People can also get an informal education by reading many books from a library or educational websites. This may also be called self-education. Some famous people have been largely self-educated, like Alfred Russell Wallace.

Distance Education is typically conducted through online platforms, correspondence courses, or

virtual classrooms, enabling flexible learning schedules and access for students who cannot attend traditional on-campus class. There are primary schools and secondary schools. In many places they are government funded. Colleges and universities usually charge fees, or tuition payments. Did you know? A developmental theory of learning was explained by psychologist Jean Piaget. It has four stages of learning through which children progress on their way to adulthood: the sensorimotor stage, pre-operational stage, concrete operational, and formal operational stages.

A teaching method is the way via which educational content is delivered by the teacher. For instance, group work is sometimes used rather than individual learning to make students grasp certain knowledge. There are many teaching methods, and the most effective one in a given (situation or) scenario depends on things like the subject matter and the learner's age and ability, or level of competence. One important part of teaching methods is to make sure that learners stay motivated, either through their own interest and curiosity or through rewards (that are outside of one's mind). This is one of my favorite things to teach! Educational assessment is the process of making documentation of the student's knowledge and skills; That can happen formally or informally and can take place before, during, or after the learning activity.

Chapter 2

Imagine we are going on an amazing adventure to learn about this! Another significant (part of teaching or) pedagogical element in many modern ways of educating, is that each lesson is part of a broad framework; The framework is (supposed to follow a syllabus or) governed by a syllabus; The syllabus can cover several months or years. According to Herbartianism, teaching can be broken down into (parts or) phases. The first phase has to do with making the student's mind ready for new information. Subsequently (or thereafter), new ideas are (shown or) introduced to the learner and then linked to (ideas or) concepts that are already familiar to them.. Special education – education of students who have a disability

Imagine we are going on an amazing adventure to learn about this! Studypool tips Archived 2024-03-02 at the Wayback Machine Let me tell you a fascinating story. Knowledge is the collection of facts, skills, and understanding that people gain through learning, experience, thinking, or discovery. It can come from things we are taught, things we figure out ourselves, or things we observe in the world. Some knowledge is easy to explain, like facts in textbooks or math formulas, this is called explicit knowledge. Other knowledge is harder to put into words, like how to ride a bike or

perform surgery, this is called tacit knowledge. It lives in our actions, habits, and personal experiences. In philosophy, knowledge has often been defined as “justified true belief.” This means that for someone to know something, they must believe it, have good reasons for believing it, and it must actually be true. However, some problems, like the “Gettier cases”, show that this definition might not always work. These cases suggest someone can have a belief that is true and has reasons behind it, but still not really have knowledge. Because of this, other ideas like reliabilism (trusting reliable ways of knowing) and virtue epistemology (focusing on intellectual character) were developed to better explain what knowledge really is. There are different kinds of knowledge. Propositional knowledge is “knowing that” something is true, like knowing that water boils at 100°C. Procedural knowledge is “knowing how” to do something, like solving a puzzle or coding. Experiential knowledge is “knowing by experience,” like recognizing a friend’s voice or knowing how a certain food tastes. Each of these kinds uses different parts of the brain and different types of thinking. The study of knowledge is called epistemology. It looks at questions like “Where does knowledge come from?” and “What can we really know?” Different thinkers have different ideas. Empiricists believe knowledge comes from what we see, hear, and touch. Rationalists believe we are born with some ideas and use reason to gain knowledge. Constructivists think we build knowledge ourselves through experience, culture, and learning from others.

Knowledge grows and improves through careful research methods like the scientific method. This includes testing ideas, sharing results with other scientists (peer review), and checking if experiments can be repeated. These steps help make sure knowledge is reliable. But in areas like art or ethics, people may have different opinions or values, so knowledge can be more personal or open to interpretation. Societies keep and share knowledge in many ways. Universities, libraries, and the internet store a huge amount of information. Other ways, like storytelling, traditions, or learning from elders, are especially important in cultures that do not rely as much on writing. These methods help pass down useful knowledge from one generation to the next. In information science, there’s a difference between data, information, and knowledge. Data is raw facts with no meaning, like numbers on a screen. Information is when that data is organized to make sense. Knowledge is when the information is put into context so that it can help someone make a decision or take action. For example, knowing it’s 35°C outside is information; understanding that it is too hot to play outside for long without water is knowledge. Neuroscience has found that knowledge is stored in networks of neurons. The cortex and hippocampus are two key areas that help us remember things. Sleep, emotions, and how we learn something can all affect how well we remember and use knowledge later. For example, if something is exciting or emotional, you are more likely to remember it. In artificial intelligence (AI), teaching computers to “know” things is a big challenge. Old systems used strict

rules to solve problems, like expert systems. Newer models, like neural networks, try to imitate how human brains work by learning patterns and relationships in data. However, even the best AI today does not really understand things like people do. It cannot think about its own thoughts or reflect on what it knows. This is one of my favorite things to teach! Knowledge can belong to individuals or groups. In companies or teams, knowledge management is used to organize what people know and help everyone share and use that knowledge well. Techniques like knowledge audits and communities of practice help keep useful skills from being lost when people leave or retire. Knowledge is not always neutral or free. In history, powerful groups have tried to control what others are allowed to know. For example, the Catholic Church once banned books that said the Earth moves around the Sun. Today, algorithms on social media can create “echo chambers” where people only see information they agree with. This shows that knowledge can be used for power, control, or influence.

Knowledge in religion is different in that it depends on faith, belief and the authority of religious leaders, not on evidence of a scientific or legal kind. There are differing views on whether religious statements should be regarded as knowledge. Close your eyes and picture this... In many expressions of Christianity, such as Catholicism and Anglicanism, knowledge is one of the seven gifts of the Holy Spirit. In the Garden of Eden knowledge is the factor that made humans greedy and treacherous. But in the Book of Proverbs it states: 'to be wise you must first obey the LORD' (9:10).

Let me tell you a fascinating story. In Islam, knowledge has great significance. "The All-Knowing" (al-⁺ô Á+m) is one of the Names of God, reflecting distinct properties of God in Islam. The Qur'an asserts that knowledge comes from God [2:239] and various hadith encourage getting knowledge. Muhammad is reported to have said "Seek knowledge from the cradle to the grave" and "Verily the men of knowledge are the inheritors of the prophets". Islamic scholars, theologians and jurists are often given the title alim, meaning 'knowledgeable'. You did an incredible job! We learned so much today. I'm very proud of you!

